Matrix Algebra!

rank of A is # of leading 1's in rref(A)

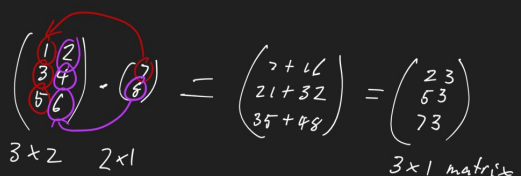
You can add & subtract & mult/divide matrices entry wise.

If A (any): $n \times m$ mat & $\vec{v} = (b_j)$: m-dim'l vector $\longrightarrow A\cdot\vec{v}$ is an n-dim vector.

$$A\cdot\vec{v} = \begin{pmatrix} a_{11}b_1 + a_{12}b_2 \cdots a_{1m}b_m \\ a_{21}b_1 + a_{22}b_2 \cdots a_{2m}b_m \\ \vdots \\ a_{n1}b_1 + a_{n2}b_2 + \cdots + a_{nm}b_m \end{pmatrix}$$

$$= \begin{pmatrix} \sum_{j=1}^{m} a_{1j}b_j \\ \sum_{j=1}^{m} a_{2j}b_j \\ \sum_{j=1}^{m} a_{nj}b_j \end{pmatrix} = \left(\sum_{j=1}^{m} a_{ij}b_j\right) \text{ for } 1 \le i \le n$$

ex:



$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix} \cdot \begin{pmatrix} 7 \\ 8 \end{pmatrix} = \begin{pmatrix} 7+16 \\ 21+32 \\ 35+48 \end{pmatrix} = \begin{pmatrix} 23 \\ 53 \\ 73 \end{pmatrix}$$

3×2 $\quad$ 2×1 $\quad$ 3×1 matrix

a system of linear equations can be modeled like this

$$\begin{aligned} x_1 + 2x_2 &= 5 \\ 3x_1 + 4x_2 &= 6 \end{aligned} \iff \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$$

Prop Let A: $n\times m$ matrix, $\vec{x}, \vec{y} \in \mathbb{R}^m$, then.

1. $A(\vec{x}+\vec{y}) = A\vec{x} + A\vec{y}$
2. $A(c\vec{x}) = c(A\cdot\vec{x})$

$\Big\}$ $A\cdot x$ is a linear operation.

Proof 1: $A(a_{ij})$, $\vec{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_m \end{pmatrix}$, $\vec{y} = \begin{pmatrix} y_1 \\ \vdots \\ y_m \end{pmatrix}$. Then

$$A\cdot(\vec{x}+\vec{y}) = A\cdot\begin{pmatrix} x_1+y_1 \\ \vdots \\ x_m+y_m \end{pmatrix} = \begin{pmatrix} \sum_{j=1}^{m} a_{1j}(x_j+y_j) \\ \vdots \\ \sum_{j=1}^{m} a_{nj}(x_j+y_j) \end{pmatrix}$$

$$A\cdot\vec{x} + A\cdot\vec{y} = \begin{pmatrix} \sum_{j=1}^{m} a_{1j}x_j \\ \vdots \\ \sum_{j=1}^{m} a_{nj}x_j \end{pmatrix} + \begin{pmatrix} \sum_{j=1}^{m} a_{1j}+y_j \\ \vdots \\ \sum_{j=1}^{m} a_{nj}+y_j \end{pmatrix}$$

$$= \begin{pmatrix} \sum_{j=1}^{m} a_{1j}(x_j+y_j) \\ \vdots \\ \sum_{j=1}^{m} a_{nj}(x_j+y_j) \end{pmatrix}$$

hence they are equal.

Proof 2 essentially same as first

Linear transformations:

$\mathbb{R}^n \to$ n dimension vector space

$\hookrightarrow \mathbb{R}^n = \left\{ \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} : a_1, \cdots, a_n \in \mathbb{R} \right\}$ is the set of all n-dimensional vector space.

Given two sets X & Y, a func $f: X \to Y$ is a collection of assignments $f(x) \in Y$ to each $x \in X$

$f: \mathbb{R} \to \mathbb{R}$ given by $f(x) = x^2$

$f: \mathbb{R}^2 \to \mathbb{R}$ given by $f(x,y) = x^2y + \sin y + e^{-x}$

Def: A function $T: \mathbb{R}^m \to \mathbb{R}^n$ of the form $T(\vec{x}) = A\cdot\vec{x}$ for an $n\times m$ mat A is called a linear transformation

Ex: A 2×2 matrix $A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ models a linear map

$T: \mathbb{R}^2 \to \mathbb{R}^2$, $T\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\cdot\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$, therefore $T(\vec{x}) = \vec{x}$

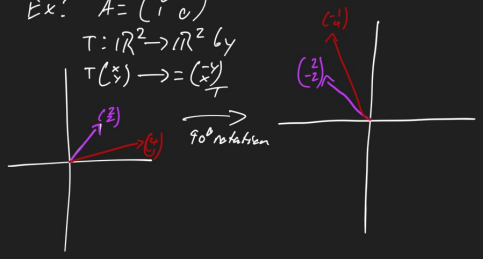
Ex: A 1×2 matrix (2×3) models a linear map

$T:\mathbb{R}^2 \to \mathbb{R}^1 (=\mathbb{R})$ by $T\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = (2\ 3)\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 2x_1 + 3x_2$

Ex: $A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$

$T:\mathbb{R}^2 \to \mathbb{R}^2$ by

$T\begin{pmatrix} x \\ y \end{pmatrix} \to = \begin{pmatrix} -y \\ x \end{pmatrix}$



Say $a_{11}x_1 + \cdots + a_{1m}x_m = b_1$

$\vdots$

$a_{n1}x_1 + \cdots + a_{nm}x_m = b_n$

$\iff$

rewrited this $A\vec{x} = \vec{b}$

$A: n\times m$, $\vec{x} \in \mathbb{R}^m$, $b = \mathbb{R}^n$

$T:\mathbb{R}^m \to \mathbb{R}^n$ by $T(\vec{x}) = A(\vec{x})$

Pick a vector $\vec{b} \in \mathbb{R}^n$

then need to find $\vec{x} \in \mathbb{R}^n$

$T(\vec{x}) = \vec{b}$

Thm let $T:\mathbb{R}^m \to \mathbb{R}^n$ be a lin.map then we can find its mat A by:

$A = \begin{pmatrix} T(\vec{e_1}) & T(\vec{e_2}) & \cdots & T(\vec{e_m}) \end{pmatrix}$ have $\vec{e_1} = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$, $\vec{e_2} = \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}$ $\cdots$ $\vec{e_m} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}$

Proof Since T is a lin.map, by def there exists $A: n\times m$ s.t. $T(\vec{x}) = A\vec{x}$

$$A = \begin{pmatrix} a_{11} & \cdots & a_{1m} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nm} \end{pmatrix} = (a_{ij})$$

$$T(\vec{e_1}) = \begin{pmatrix} a_{11} & \cdots & a_{1m} \\ \vdots & & \vdots \\ a_{n1} & \cdots & a_{nm} \end{pmatrix}\begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} = \begin{pmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{n1} \end{pmatrix} \qquad T(\vec{e_2}) = \begin{pmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{n2} \end{pmatrix}$$

ex: $T:\mathbb{R}^2 \to \mathbb{R}^2$ given by 90° rotation

assume T is a lin.map what is the mat A defining T.



$T\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, $T\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \end{pmatrix}$

$A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$

for 2×2 $e_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ $e_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

thm let $T:\mathbb{R}^m \to \mathbb{R}^n$ be a function

the following hold:

1. $T(\vec{u}+\vec{v}) = T(\vec{u}) + T(\vec{v})$ for all $\vec{u}, \vec{v} \in \mathbb{R}^m$
2. $T(c\vec{v}) = cT(\vec{v})$ for all $\vec{v} \in \mathbb{R}^m$, $c \in \mathbb{R}$

def a function $T:\mathbb{R}^m \to \mathbb{R}^n$ is a lin.map if

1. $T(\vec{u}+\vec{v}) = T(\vec{u}) + T(\vec{v})$ for all $\vec{u}, \vec{v} \in \mathbb{R}^m$
2. $T(c\vec{v}) = cT(\vec{v})$ for all $\vec{v} \in \mathbb{R}^m$, $c \in \mathbb{R}$